

Reciprocal Sums and the Sylvester Recurrence

Erdős Problem #243

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ABSTRACT

We prove irrationality of the reciprocal sum when strictly increasing positive integers satisfy $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$. The proof extracts a cubic polynomial from an integer tail numerator, then uses a square in a cubic field to restrict its coefficients. Congruences modulo seven exclude the remaining possibilities. We also obtain Sylvester recurrence criteria from bounded numerator increases and weighted sums over new maxima. The unrestricted Erdős–Graham question remains open.

1 Introduction

The Sylvester recurrence $a_{n+1} = a_n^2 - a_n + 1$ gives a rational reciprocal sum by telescoping:

$$\frac{1}{a_n - 1} = \frac{1}{a_n} + \frac{1}{a_{n+1} - 1}.$$

For an increasing sequence of positive integers that eventually satisfies this recurrence, the tail sum at every sufficiently large index is $1/(a_n - 1)$. Erdős and Graham asked whether every rational reciprocal sum with $a_{n+1} \sim a_n^2$ arises in this way.

Problem 1.1 (Erdős Problem 243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

Erdős and Graham [5, p. 64] asked this question; see also Erdős [6, p. 105] and Bloom’s catalogue [11]. We prove the following irrationality result for a restricted class of sequences with $a_{n+1} \sim a_n^2$.

Theorem 1.2 (cubic-rate irrationality). *A strictly increasing sequence of positive integers with*

$$a_n^2/a_{n+1} = 1 + \frac{3}{n} + o(n^{-3})$$

has irrational reciprocal sum.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Under the eventual Sylvester recurrence, $a_n^2/a_{n+1} - 1 = O(1/a_n)$, which is much smaller than $3/n$. Theorem 1.2 therefore treats a restricted case of the problem by excluding rationality altogether. The general quadratic limit $a_{n+1} \sim a_n^2$ remains insufficient for any argument given here.

For example, set $a_1 = 8$ and

$$a_{n+1} = \left\lceil \frac{na_n^2}{n+3} \right\rceil.$$

This gives 8, 16, 103, 5305, From $a_{n+1} \geq a_n^2/4$ we obtain $a_n \geq 4 \cdot 2^{2^{n-1}}$ by induction. The error introduced by the ceiling is less than one, and hence

$$0 \leq 1 + \frac{3}{n} - \frac{a_n^2}{a_{n+1}} < \frac{16}{a_n^2} = o(n^{-3}).$$

The theorem applies to this explicitly defined sequence.

If the sum is rational, clearing denominators gives positive integer tail numerators C_n . Their growth is cubic, so we compare them with $F_n = n(n+1)(n+2)$, for which $F_{n+1}/F_n = 1+3/n$. The ratio estimate gives $C_n = AF_n + o(n)$. Subtracting the recurrences for C_n and AF_n then shows that the first difference of the error tends to zero. It follows that $\Delta^4 C_n \rightarrow 0$. Since these fourth differences are integers, they eventually vanish, and $C_n = AF_n + B$.

We next use the denominator recurrence to constrain this polynomial. It gives a square condition at every root of a cubic modulo almost every prime. Chebotarev's theorem implies that the corresponding element of the cubic number field is a square. Trace identities leave two possible polynomials; four consecutive terms modulo seven exclude both. The proof in Section 3 follows this order.

For the recurrence criteria we use the integer tail numerators of Koizumi [10, Lemma 4]. In Section 4, a stable gcd and the Chinese remainder theorem exclude a rational tail whose quotient $a_1 \cdots a_{n-1}/a_n$ has bounded upward increments. An elementary infinite product argument gives the separate summability criterion of Section 5. We then clear the tails by a least common multiple and count crossings of new maxima (Section 6). The resulting weighted criterion isolates an arithmetic estimate still missing from the general problem. The companion paper also treats nonintegral rates and proves a stronger cubic disagreement theorem. Its detailed proof locations and the formal sources are listed in Appendix B.

2 Tail numerators

For the Sylvester sequence 2, 3, 7, 43, ..., whose reciprocal sum is 1, the successive tails are $1, 1/2, 1/6, 1/42, \dots$. Multiplication by the products $1, 2, 6, 42, \dots$ leaves the constant integer numerator 1. For an arbitrary rational reciprocal sum, we also multiply by a denominator of the whole sum. The resulting integer numerators need not be constant.

Assume now that a_n are strictly increasing positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n = p/q$ with positive integers p, q . All indices in this construction start at 1.

Put

$$x_n = \sum_{k \geq n} \frac{1}{a_k}, \quad P_n = \prod_{1 \leq j < n} a_j, \quad D_n = qP_n, \quad C_n = D_n x_n.$$

Here $P_1 = 1$, $D_1 = q$ and $C_1 = p$. Since every a_k with $k < n$ divides P_n , subtracting the finite prefix from p/q gives

$$C_n = pP_n - q \sum_{1 \leq k < n} \frac{P_n}{a_k} \in \mathbb{N}_{>0}.$$

The integer is positive because it equals the positive tail multiplied by D_n . Removing the term $1/a_n$ from the tail gives

$$C_{n+1} = a_n C_n - D_n, \quad D_{n+1} = a_n D_n. \quad (1)$$

Koizumi constructs these coordinates in Lemma 4 [10, pp. 11–12]. On a pseudo-greedy tail his c_n, d_n, e_n correspond to $C_n, D_n, D_n - (a_n - 1)C_n$. If the tail is restarted with its value in lowest terms, both coordinates are divided by the same fixed integer. We retain the unreduced fraction C_n/D_n to keep the denominator update multiplicative. In the cubic proof we first extract a polynomial for this C_n , then divide out the eventual fixed gcd in Section 3.2.

For large n , $a_{n+1} \geq a_n^2/2 \geq 2a_n$. The terms after $1/a_{n+1}$ sum to at most $2/a_{n+2} \leq 4/a_{n+1}^2$, so

$$x_n = \frac{1}{a_n} + \frac{1}{a_{n+1}} + O(a_{n+1}^{-2}), \quad \frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} + O(1/a_n). \quad (2)$$

For the second estimate, the first gives $a_n x_n = 1 + O(1/a_n)$. Use this in

$$\frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} \frac{a_{n+1} x_{n+1}}{a_n x_n}.$$

The last factor is $1 + O(1/a_n)$, and $a_n^2/a_{n+1} \rightarrow 1$, giving the stated additive error. Quadratic growth makes $1/a_n = o(n^{-k})$ for every fixed k , so this error preserves the precision required in the cubic theorem. We keep the original index n throughout that proof. Although an eventual recurrence survives removal of a finite prefix, the expression $1 + 3/n + o(n^{-3})$ changes its lower-order terms under an index shift.

3 Proof of cubic-rate irrationality

Suppose, towards a contradiction, that the reciprocal sum in Theorem 1.2 is rational. Construct C_n, D_n as in Section 2. Equation (2) transfers the rate without changing its precision:

$$\frac{C_{n+1}}{C_n} = 1 + \frac{3}{n} + o(n^{-3}). \quad (3)$$

We first show that C_n is eventually a polynomial, using (3) and integrality. We then apply (1) to restrict its coefficients.

3.1 An eventual cubic polynomial

The factor $1 + 3/n$ suggests the cubic $F_n = n(n+1)(n+2)$, since $F_{n+1}/F_n = 1 + 3/n$. Write (3) as $C_{n+1} = (1 + 3/n + \varepsilon_n)C_n$, where $\varepsilon_n = o(n^{-3})$. Dividing out the exact model gives

$$z_n = \frac{C_n}{F_n}, \quad \frac{z_{n+1}}{z_n} = 1 + \frac{\varepsilon_n}{1 + 3/n}.$$

The logarithmic increments are absolutely summable, with tails $o(n^{-2})$. Hence z_n tends to some $A > 0$ and $z_n - A = o(n^{-2})$. Multiplying back by F_n yields

$$C_n = AF_n + \delta_n, \quad \delta_n = o(n).$$

The estimate $\delta_n = o(n)$ alone does not control its differences. Instead, subtracting the recurrences for C_n and AF_n gives

$$\Delta\delta_n = \frac{3}{n}\delta_n + \varepsilon_n C_n = o(1), \quad \Delta h_n = h_{n+1} - h_n.$$

Here $C_n = O(n^3)$ makes the second term $o(1)$. Three more differences still tend to zero and kill the cubic model:

$$\Delta^4 C_n = \Delta^3(\Delta\delta_n) \rightarrow 0.$$

The left side is an integer, so it vanishes for all sufficiently large n . Newton interpolation therefore expresses C_n on that tail as a rational polynomial of degree at most three. Its difference from AF_n is $o(n)$, which permits only a constant term:

$$C_n = An(n+1)(n+2) + B, \quad A \in \mathbb{Q}_{>0}, \quad B \in \mathbb{Q}. \quad (4)$$

The precision matters at the subtraction step. The integer sequence $F_n + (-1)^n$ has ratio $1 + 3/n + O(n^{-3})$ and never agrees eventually with a polynomial; its error has an oscillating first difference. This example concerns polynomial extraction alone. It is not asserted to satisfy the denominator recurrence.

3.2 Reduction and the constant term

We now use the denominator recurrence. Both updates preserve common divisors, so $G_n = \gcd(C_n, D_n)$ forms a divisibility chain: $G_n \mid G_{n+1}$. For every sufficiently late n , it divides four consecutive values of C_n , and hence their third difference $6A$. Unlike the vanishing fourth difference used above, this positive third difference bounds the gcd. Thus G_n stabilises at a positive integer g .

We can therefore reduce all sufficiently late fractions by the same factor: put $u_n = C_n/g$ and $v_n = D_n/g$. The recurrences persist after this division. The following observation describes the resulting coprimality conditions.

Proposition 3.1 (persistent coprimality). *Let a_n, u_n, v_n be integer sequences with $u_n > 0$, $v_n \geq 0$ and $\gcd(u_n, v_n) = 1$ satisfying*

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n.$$

Then $\gcd(a_n, v_n) = 1$. The a_n are pairwise coprime, and $\gcd(a_i, u_t) = 1$ whenever $i < t$.

Proof. A prime dividing both a_n and v_n would divide u_{n+1} and v_{n+1} , contrary to reducedness. For $i < t$, the factor a_i divides v_t . Reducedness gives $\gcd(a_i, u_t) = 1$, and $\gcd(a_t, v_t) = 1$ gives $\gcd(a_i, a_t) = 1$. $\square$

In the cubic case, adjacent numerators are also coprime, since $\gcd(u_n, u_{n+1}) = \gcd(u_n, v_n) = 1$. Write the polynomial for u_n as

$$Q(n) = \frac{m}{6}n(n+1)(n+2) + c. \quad (5)$$

The third difference $m = 6A/g$ is a positive integer. Since $n(n+1)(n+2)/6 = \binom{n+2}{3}$ is an integer, any sufficiently late value $u_n = Q(n)$ gives

$$c = u_n - m \binom{n+2}{3} \in \mathbb{Z}.$$

Thus Q is integer-valued, although its coefficients need not all be integers. If a prime ℓ divides c , choose an arbitrarily late $n \equiv -1 \pmod{6\ell}$. Both $Q(n)$ and $Q(n+1)$ would be divisible by ℓ , contradicting adjacent coprimality. This also excludes $c = 0$. Thus

$$m \in \mathbb{N}_{>0}, \quad c \in \{1, -1\}.$$

We also need irreducibility. The multipliers $a_n > 1$ on the reduced tail are pairwise coprime, so infinitely many distinct primes occur among them. Once a prime divides one such a_n , it divides every later v_t and no later u_t . If the cubic Q had a rational root, that root would reduce to a root modulo all but finitely many primes. Choose one of the persistent primes outside those exceptions; a sufficiently late index in the root's residue class would have $\ell \mid Q(t) = u_t$, a contradiction. A cubic is reducible over $\mathbb{Q}$ only if it has a rational root. Hence Q is irreducible.

Set $\kappa = m/6$, $\eta = 6c/m$, and

$$f(T) = T^3 - T + \eta, \quad Q(n) = \kappa f(n+1).$$

3.3 A square forced by a zero numerator

A zero numerator modulo a prime eliminates one term of the recurrence. To use this, first eliminate v_n from the two exact updates:

$$u_{n+2} = (a_n + a_{n+1})u_{n+1} - a_n^2 u_n. \quad (6)$$

Whenever $u_k \equiv 0 \pmod{\ell}$, it follows that

$$-u_{k-1}u_{k+1} \equiv (a_{k-1}u_{k-1})^2 \pmod{\ell}.$$

Thus the negative product of the neighbouring numerators must be a square. Since our numerator is a polynomial, its roots modulo ℓ tell us exactly where to apply this observation.

Let $\ell \nmid 6m$ be prime and let $r \in \mathbb{F}_\ell$ satisfy $f(r) = 0$. The constant $\eta = 6c/m$ is nonzero modulo ℓ , so $r \notin \{0, 1, -1\}$. Choose a sufficiently late k with $k+1 \equiv r \pmod{\ell}$. Then $u_k = \kappa f(k+1) \equiv 0$, whereas

$$f(r-1) = -3r(r-1), \quad f(r+1) = 3r(r+1).$$

Consequently, in $\mathbb{F}_\ell$,

$$(a_{k-1}u_{k-1})^2 = -u_{k-1}u_{k+1} = 9\kappa^2 r^2 (r^2 - 1).$$

Dividing by the nonzero square $9\kappa^2 r^2$ gives

$$\ell \nmid 6m, \quad r \in \mathbb{F}_\ell, \quad f(r) = 0 \quad \implies \quad r^2 - 1 \in (\mathbb{F}_\ell^\times)^2 \quad (\ell \text{ prime}). \quad (7)$$

Every root is covered. A prime at which f has no root imposes no condition.

Let α be a root of f and put $K = \mathbb{Q}(\alpha)$, a cubic field. We claim that $\alpha^2 - 1$ is a square in K . A nonsquare would give a quadratic extension in which an automorphism fixes α but changes the sign of its square root. Chebotarev realises this behaviour modulo a prime: the reduced α lies in the prime field, while the square root does not. That would contradict (7).

Here are the details of this passage. Suppose $\alpha^2 - 1$ is not a square in K , and choose β with $\beta^2 = \alpha^2 - 1$. The nontrivial automorphism of $K(\beta)/K$ fixes α and sends β to $-\beta$. Extend it to an element $\sigma \in \text{Gal}(M/\mathbb{Q})$, where M is a Galois closure of $K(\beta)/\mathbb{Q}$.

Chebotarev's density theorem [9, §3, author-version p. 15] supplies infinitely many unramified rational primes whose Frobenius is conjugate to σ . Choosing a prime of M above each of them makes the Frobenius equal to σ itself. Omit residue characteristic two, primes dividing $6m$, and the finitely many primes where α or β is nonintegral or β reduces to zero. At a remaining prime, Frobenius acts by the ℓ -th power on the residue field, so

$$\bar{\alpha}^\ell = \bar{\alpha}, \quad \bar{\beta}^\ell = -\bar{\beta} \neq \bar{\beta}.$$

The first equality puts $r = \bar{\alpha}$ in $\mathbb{F}_\ell$ and gives $f(r) = 0$. The second keeps both roots $\pm\bar{\beta}$ of $X^2 - (r^2 - 1)$ outside $\mathbb{F}_\ell$. This contradicts (7), which applies to the particular root selected by Frobenius. Removing finitely many primes has left infinitely many choices. Hence there is a $\beta \in K$ with $\beta^2 = \alpha^2 - 1$.

3.4 Traces in the cubic field

To constrain η , we use the square relation to express α through a pair of reciprocal elements in the same cubic field:

$$(\alpha + \beta)(\alpha - \beta) = 1.$$

Put $z = \alpha + \beta$. Then $z^{-1} = \alpha - \beta$ and $2\alpha = z + z^{-1}$. Since $z \in K$ and $\alpha \in \mathbb{Q}(z)$, we have $\mathbb{Q}(z) = K$. Write the minimal polynomial of z as

$$p(T) = T^3 + bT^2 + dT + w, \quad b, d, w \in \mathbb{Q}, \quad w \neq 0.$$

The minimal polynomial of z^{-1} is the monic reciprocal polynomial

$$\frac{T^3 p(T^{-1})}{w} = T^3 + \frac{d}{w}T^2 + \frac{b}{w}T + \frac{1}{w}.$$

These two polynomials give the traces of z and z^{-1} by the same Newton identities. All traces below are from K to $\mathbb{Q}$.

The cubic for α determines its first three traces:

$$\mathrm{Tr}(\alpha) = 0, \quad \mathrm{Tr}(\alpha^2) = 2, \quad \mathrm{Tr}(\alpha^3) = -3\eta.$$

Expressing these same traces through z and its reciprocal gives

$$d = -bw, \quad b^2 + b(w - w^{-1}) = 1, \quad \eta = \frac{(b^2 + 1)(w + w^{-1})}{8}. \quad (8)$$

For completeness, the first equality follows from $0 = 2 \mathrm{Tr}(\alpha) = -b - d/w$. After substituting $d = -bw$, Newton's identities give

$$\begin{aligned} \mathrm{Tr}(z^2) + \mathrm{Tr}(z^{-2}) &= 2b^2 + 2b(w - w^{-1}), \\ \mathrm{Tr}(z^3) + \mathrm{Tr}(z^{-3}) &= -3(b^2 + 1)(w + w^{-1}). \end{aligned}$$

The first line is $4 \mathrm{Tr}(\alpha^2) - 6 = 2$. The second is $8 \mathrm{Tr}(\alpha^3) = -24\eta$, because the mixed terms in $(z + z^{-1})^3$ have trace $3 \mathrm{Tr}(z + z^{-1}) = 0$. These give the other two equalities in (8).

Multiplying the middle equation by w and bringing all terms to one side gives $(bw - 1)(b + w) = 0$. Write $w = r/s$ in lowest terms, with $r \neq 0, s > 0$. Substituting $b = -w$ or $b = 1/w$ into the last equation and using $\eta = 6c/m$ gives respectively

$$m = \frac{48c rs^3}{(r^2 + s^2)^2} \quad \text{or} \quad m = \frac{48c r^3 s}{(r^2 + s^2)^2}.$$

Since $\gcd(r^2 + s^2, rs) = 1$ and $m \in \mathbb{Z}$, the square $(r^2 + s^2)^2$ divides 48. Hence $r^2 + s^2$ is 1, 2 or 4. Both r and s are nonzero, which leaves only 2: a sum of two nonzero integer squares cannot equal 1 or 4. Thus $|r| = s = 1$, and positivity of m gives $m = 12$. The only remaining numerators are

$$Q(n) = 2n(n+1)(n+2) + 1 \quad \text{and} \quad Q(n) = 2n(n+1)(n+2) - 1.$$

3.5 The contradiction modulo seven

In the plus case, choose a late block starting at $n \equiv 0 \pmod{7}$; in the minus case, start at $n \equiv 1 \pmod{7}$. The four successive numerators have residues

$$(u_n, u_{n+1}, u_{n+2}, u_{n+3}) \equiv \begin{cases} (1, 6, 0, 2) & (c = 1), \\ (4, 5, 0, 1) & (c = -1). \end{cases}$$

Call the first two denominator residues d_0, d_1 and the displayed numerator residues $c_0, c_1, 0, c_3$. From $0 = a_{n+1}c_1 - d_1$ and $c_3 = -a_{n+1}d_1$ we get $d_1^2 = -c_1c_3 = 2$. Thus $d_1 \in \{3, 4\}$. The first update, however, gives

$$d_1 = a_n d_0 = \frac{d_0(d_0 + c_1)}{c_0}.$$

Completing the square in these two cases gives, in $\mathbb{F}_7$,

$$d_1 + 2 = \begin{cases} (d_0 + 3)^2 & (c = 1), \\ 2(d_0 - 1)^2 & (c = -1). \end{cases}$$

Since $2 = 3^2$ in this field, $d_1 + 2$ must be a square in either case. But $d_1 \in \{3, 4\}$ would give $d_1 + 2 \in \{5, 6\}$, both nonsquares. The middle zero and its two neighbours supplied $d_1^2 = 2$; the preceding update supplied the incompatible second square. All four numerators are used. Both cubics are excluded, proving Theorem 1.2.

Section 2 of the companion proves positive lower density of disagreement with each rational cubic of this form. That lower bound depends on the orbit and the cubic. For Theorem 1.2, exclusion of eventual equality suffices.

4 Bounded increases

We return to the rational tails of Section 2. The Sylvester example had a constant integer numerator. To measure departure from this behaviour, put

$$E_n = D_n - (a_n - 1)C_n = C_n - C_{n+1}.$$

Thus a negative error is an upward step of C_n , and zero error says that the reciprocal tail equals $1/(a_n - 1)$. The next theorem forces constancy from bounded upward steps and a vanishing relative error. The latter limit already follows from quadratic growth; the bound on upward steps is an additional hypothesis. No polynomial expansion is assumed.

Theorem 4.1 (bounded negative part). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $a_n > 1$, $C_n > 0$, and*

$$C_{n+1} + D_n = a_n C_n, \quad D_{n+1} = a_n D_n \quad (n \geq 0).$$

Put $E_n = D_n - (a_n - 1)C_n$. If

$$\frac{E_n}{C_n} \rightarrow 0 \quad \text{and} \quad E_n \geq -B \quad \text{for all sufficiently large } n$$

for some integer $B \geq 0$, then $E_n = 0$ for all sufficiently large n .

The relative limit is equivalently the assertion that, for every $K \in \mathbb{N}$, we eventually have $K|E_n| < C_n$. This is the division-free form used in the linked theorem. The case $K = 1$ lets us propagate a zero error. In the absence of late zeros, the same limit and integrality force $C_n \rightarrow \infty$. We shall then use the bound on upward steps to stabilise the gcd and produce a block of consecutive integers that the numerator cannot cross.

4.1 Zero errors and descent

Throughout this section, $E_n = D_n - (a_n - 1)C_n$ and $z_+ = \max(z, 0)$. The arguments below use the recurrences (1). We state positivity and the bound $|E_n| < C_n$ where they are needed.

Proposition 4.2 (error identities). *For integer sequences satisfying (1),*

$$C_{n+1} = C_n - E_n, \quad (a_{n+1} - a_n^2 + a_n - 1)C_{n+1} = a_n^2 E_n - E_{n+1}.$$

Proof. The first identity follows by substituting $D_n = E_n + (a_n - 1)C_n$ into the numerator recurrence. Substituting this expression at the next index and using $D_{n+1} = a_n D_n$ gives the second. $\square$

Equation (2) now gives $E_n/C_n = 1 - C_{n+1}/C_n \rightarrow 0$, so $|E_n| < C_n$ for all sufficiently large n . Koizumi proves this relative-gap limit for an eventual pseudo-greedy tail [10, Corollary 3, p. 9]. His rounding convention also gives $-C_n/2 \leq E_n < C_n/2$, a stronger bound than we need.

Koizumi's Lemma 3 gives zero absorption for pseudo-greedy tails, and Proposition 1(2) uses descent when their gaps are eventually nonnegative [10, p. 10 and p. 14]. The following form separates what the two arguments require.

Theorem 4.3 (absorption and descent). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $a_n > 1$, $C_n > 0$ and (1). If $|E_n| < C_n$ for every n , then $E_n = 0$ implies $E_{n+1} = 0$. For any positive integer sequence C_n with $C_{n+1} = C_n - E_n$, eventual nonnegativity of E_n implies its eventual vanishing.*

Proof. When $E_n = 0$, the second identity in Proposition 4.2 shows that C_{n+1} divides E_{n+1} . The inequality $|E_{n+1}| < C_{n+1}$ then forces $E_{n+1} = 0$. For the second assertion, $C_{n+1} = C_n - E_n$ makes the positive integer sequence C_n eventually nonincreasing, hence constant. $\square$

Corollary 4.4 (two zero errors). *For integer sequences satisfying (1), if $E_n = E_{n+1} = 0$ and $C_{n+1} \neq 0$, then $a_{n+1} = a_n^2 - a_n + 1$. In particular, if $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ also satisfy $a_n > 1$ and $C_n > 0$, eventual zero error implies the eventual Sylvester recurrence.*

Proof. Cancel the nonzero factor C_{n+1} in the second error identity. $\square$

Once $|E_n| < C_n$ holds at every later index, one zero makes every subsequent error zero. If there are no late zeros, integrality gives $|E_n| \geq 1$, and $|E_n|/C_n \rightarrow 0$ forces $C_n \rightarrow \infty$. The scalar update alone permits $C_n = n + 1$, $E_n = -1$ forever. To exclude such behaviour we shall use the denominator recurrence.

4.2 A stable gcd and a forbidden block

Proposition 4.5 (gcd stabilisation). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $a_n > 1$, $C_n > 0$ and (1). If some fixed integer $B \geq 1$ satisfies $-B \leq E_n < 0$ at infinitely many indices, then $G_n = \gcd(C_n, D_n)$ is eventually constant. Dividing C_n, D_n by its eventual value gives coprime sequences satisfying the same recurrences.*

Proof. We have $G_n \mid G_{n+1}$ and $G_n \mid E_n$. Given n , choose $t \geq n$ with $-B \leq E_t < 0$. Then $G_n \leq G_t \leq -E_t \leq B$. Hence the positive divisibility chain stabilises. Division by its stable value preserves the recurrences and gives coprime numerators and denominators. $\square$

For this gcd argument, bounded negative errors need occur only arbitrarily late. To control the first crossing below, we shall need the upper bound on every sufficiently late upward step.

A sequence avoiding multiples of 2 and 3 cannot cross the pair $6k + 2, 6k + 3$ from below with upward steps at most two: its first value at least $6k + 2$ would be one of those two integers. Decreases before that crossing do not help. For a general step bound B , the Chinese remainder theorem supplies a forbidden block of length B .

Lemma 4.6 (consecutive multiples). *For pairwise coprime integers $m_0, \dots, m_{B-1} \geq 2$ and every lower bound, there is a larger t such that $m_i \mid t + i$ for each $i < B$.*

Proof. Solve $t \equiv -i \pmod{m_i}$ by the Chinese remainder theorem and add multiples of $\prod_i m_i$ to exceed the prescribed bound. $\square$

Theorem 4.7 (Chinese remainder theorem and first crossing). *Let $u : \mathbb{N} \rightarrow \mathbb{N}$ tend to infinity and let $B \geq 1$ be an integer with $u_{n+1} \leq u_n + B$ for every n . There is no sequence of pairwise coprime integers $m_i \geq 2$ for which $\gcd(m_i, u_t) = 1$ whenever $i < t$.*

Proof. Apply Lemma 4.6 to $m_0, \dots, m_{B-1}$, choosing $t > \max(u_0, \dots, u_B)$. There is a first $n > B$ with $u_n \geq t$. Then

$$t \leq u_n \leq u_{n-1} + B < t + B.$$

Thus $u_n = t + i$ for some $i < B < n$, and $m_i \mid u_n$, contrary to $\gcd(m_i, u_n) = 1$. $\square$

Bado proves a two-sided bounded-error criterion by this forbidden-block argument [7, Theorem 5.1, pp. 4–5]. The gcd stabilisation above allows us to use only the lower error bound.

Proof of Theorem 4.1. Suppose that the conclusion fails. Absorption excludes every sufficiently late zero. Since $|E_n| \geq 1$ there, the relative limit gives $C_n \rightarrow \infty$. There must also be infinitely many negative errors, for otherwise integer descent would make C_n constant. At those late negative indices, $1 \leq -E_n \leq B$, so $B \geq 1$ and Proposition 4.5 gives a stable gcd g . We divide by g and obtain $u_n = C_n/g \rightarrow \infty$ with

$$u_{n+1} - u_n = -E_n/g \leq B.$$

By Proposition 3.1, the multipliers $a_i \geq 2$ are pairwise coprime and each is coprime to every later numerator. They supply the moduli prohibited by Theorem 4.7, a contradiction. $\square$

Corollary 4.8. *Under Theorem 4.1, the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ eventually.*

Proof. Theorem 4.1 gives eventual zero error; positivity of C_n and Corollary 4.4 give the recurrence. $\square$

4.3 The condition in the original sequence

Let $\gamma_n = a_n^2/a_{n+1} - 1$. For $t_n = P_n/a_n$,

$$t_{n+1} - t_n = t_n \gamma_n.$$

Thus the following hypothesis bounds upward increments of t_n . It allows t_n itself to be unbounded and allows decreases.

Corollary 4.9 (bounded increments of P_n/a_n). *Let $a_1 < a_2 < \dots$ be positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n \in \mathbb{Q}$. Put $P_n = \prod_{j < n} a_j$. If*

$$\limsup_{n \rightarrow \infty} \frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < +\infty,$$

then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .

Proof. For the canonical integer tail, direct substitution gives

$$E_n + q \frac{P_n}{a_n} \gamma_n = q P_n \left(\frac{1}{a_{n+1}} - (a_n - 1) \sum_{k \geq n+2} \frac{1}{a_k} \right). \quad (9)$$

By the tail estimate, the subtracted term is at most $4(a_n - 1)/a_{n+1}^2 < 1/a_{n+1}$ eventually. The right side is positive, so an upper bound on $(P_n/a_n)\gamma_n$ gives a lower bound on E_n . The tail construction supplies positivity and the exact recurrences, while (2) gives $E_n/C_n \rightarrow 0$. Theorem 4.1 makes the error eventually zero, and Corollary 4.8 gives the Sylvester recurrence. $\square$

Koizumi's proof of Corollary 4(1) uses this product-weighted comparison [10, (11)–(12), p. 15]; that corollary assumes a nonpositive upper limit, whereas the argument here allows any finite upper limit. A corresponding LCM comparison appears below. For $a_n = b^{2^{n-1}}$, $b \geq 2$, the increments of P_n/a_n are zero. Since the sequence does not eventually satisfy the Sylvester recurrence, the corollary gives irrationality. For a Sylvester sequence starting at $a_1 > 1$, $a_n - 1 = (a_1 - 1)P_n$ and the increments tend to zero.

The inclusive rate $1/n$ gives another useful sufficient condition.

Corollary 4.10 (an inclusive one-sided $1/n$ bound). *Let $a_1 < a_2 < \dots$ be positive integers with $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum_n 1/a_n \in \mathbb{Q}$. Suppose that for some $K \geq 0$ and $\varepsilon > 0$,*

$$\gamma_n := a_n^2/a_{n+1} - 1 \leq \frac{1}{n} + \frac{K}{n^{1+\varepsilon}} \quad \text{for all large } n.$$

Then the sequence is eventually Sylvester. In particular, the conclusion holds under the pointwise eventual bound $\gamma_n \leq 1/n$.

Proof. Since $t_{n+1}/t_n = 1 + \gamma_n$, division by the comparison sequence n gives $(t_{n+1}/(n+1))/(t_n/n) = (1+\gamma_n)/(1+1/n) \leq 1 + K/[n^\varepsilon(n+1)]$. The upper bounds have a convergent product, so $t_n = O(n)$ and $t_n \gamma_n \leq t_n(\gamma_n)_+ = O(1)$. Corollary 4.9 applies. $\square$

The weaker condition $\limsup n(\gamma_n)_+ \leq 1$ does not suffice for the estimate $t_n = O(n)$. For example, the scalar ratios $1 + \gamma_n = 1 + (1 + 1/\log n)/n$ give a product of order $n \log n$. This example does not supply a reciprocal-tail counterexample.

For positive limiting increments, take $a_1 = 4$ and $a_{n+1} = \lceil na_n^2/(n+1) \rceil$, beginning 4, 8, 43, 1387. Here $a_n \geq 2 \cdot 2^{2^{n-1}}$ and

$$0 \leq 1 + \frac{1}{n} - \frac{a_n^2}{a_{n+1}} < \frac{4}{a_n^2}.$$

Consequently $t_n \sim Kn$ and $t_{n+1} - t_n \rightarrow K > 0$. The corollary proves irrationality because $\gamma_n \sim 1/n$ is incompatible with the eventual Sylvester recurrence.

For comparison, the ceiling example with rate $3/n$ in the introduction has $t_n \sim K_3 n^3$ and $t_{n+1} - t_n \sim 3K_3 n^2$ for some $K_3 > 0$. Indeed, divide t_n by $n(n+1)(n+2)$ and use the absolutely summable rounding errors, as in Section 3.1. The power sequence has constant t_n , the $1/n$ example has bounded positive limiting increments, and the cubic example has unbounded increments. The last therefore needs the arithmetic proof of Theorem 1.2.

5 Finite total relative increase

An infinite product gives another way to bound a numerator. This argument needs only the scalar update $C_{n+1} = C_n - E_n$ and positivity on a discrete set. We assume summability of the relative upward increments. Mere convergence of those increments to zero would allow $C_n = n + 1$ and $C_n = (n + 1)^2$, both of which grow without bound.

Theorem 5.1 (a convergent sum of relative increases). *Let C_n be positive integers and E_n integers satisfying $C_{n+1} = C_n - E_n$. If*

$$\sum_n \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. Neither denominator dynamics nor vanishing relative error is required.

Proof. Set $\delta_n = (-E_n)_+/C_n$. Then

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n) \leq C_0 \exp\left(\sum_n \delta_n\right).$$

Choose an integer K above this bound. A strict rise of the integer sequence has size at least one, so it contributes at least $1/K$ to $\sum \delta_n$. Hence only finitely many rises occur. Thereafter C_n is a nonincreasing positive integer sequence, and its eventual constancy gives $E_n = 0$. $\square$

The proof also allows positive real C_n with integer E_n : the values lie in $C_0 + \mathbb{Z}$, whose bounded positive part is finite. Discrete increments are essential to this argument. With real errors, $C_n = 1 + 1/(n + 1)$ decreases forever with zero relative-increase sum; without positivity, $C_n = -n - 1$, $E_n = 1$ does so too. The companion's Section 13 gives the details.

On an exact reciprocal-tail orbit the conclusion gives the Sylvester recurrence. For the gap sequence of the pseudo-greedy expansion, the same criterion, with the same product bound and integer descent, appears in the Erdős Problem a Day working report on Problem #243, dated 12 August 2026 [8, "A global termination criterion"]; the statement above is for any positive integer sequence with $C_{n+1} = C_n - E_n$. This summability condition and the bound in Theorem 4.1 are different estimates. For exact reciprocal tails with $|E_n|/C_n \rightarrow 0$, each is equivalent to eventual vanishing of E_n ; neither estimate has been derived here from the unrestricted problem.

6 New maxima of an LCM numerator

The product P_n counts a prime factor again each time it occurs in an earlier denominator. A least common multiple retains only its largest power, and so can give a smaller clearing denominator. We shall sum increases of the resulting integer numerator only when they create a new maximum. First crossings of selected heights will then control these sums even with a decreasing weight. Put

$$L_n = \text{lcm}(q, a_1, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = L_n x_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The denominator of x_n divides L_n , so $U_n \in \mathbb{Z}$ and $V_n = L_n - (a_n - 1)U_n \in \mathbb{Z}$. The fraction U_n/L_n need not be reduced. With $\rho_n = \gcd(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

These coordinates and updates also appear in Bado's September 2026 preprint. With his denominator parameter equal to q , his M_{n-1} , Δ_{n-1} , K_{n-1} , u_n and g_n are respectively our L_n , M_n , U_n , V_n and ρ_n ; his (19) is the middle update above [7, Proposition 7.1 and (16)–(20), pp. 6–7].

Since $L_n \mid qP_n$ and $V_n = (L_n/(qP_n))E_n$, multiplying (9) by $L_n/(qP_n)$ gives, eventually,

$$0 < V_n + \frac{L_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) \leq \frac{L_n}{a_{n+1}}. \quad (10)$$

The lower bound will transfer a growth condition to a bound on V_n . For the weighted version we will also need the upper bound, which makes the comparison error summable.

Write $R_n = \max_{j \leq n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. The inequality $|V_n| < U_n$ gives $U_{n+1} < U_n$ when $\rho_n \geq 2$, so every sufficiently late strict rise has $\rho_n = 1$. The stronger eventual bound $-U_n \leq 2V_n$, supplied by $V_n/U_n = E_n/C_n \rightarrow 0$, gives the quantitative estimate $U_{n+1} \leq 3U_n/4$ when $\rho_n \geq 2$. At a sufficiently late record step, where $\rho_n = 1$, the actual jump is $d_n = U_{n+1} - U_n = -V_n > 0$. This identity is not asserted at a contracting step with $\rho_n \geq 2$.

Theorem 6.1 (a convergent weighted sum over new maxima). *Assume the growth and rationality hypotheses of Problem 1.1. Let $f : [1, \infty) \rightarrow [0, \infty)$ be finite and nonincreasing, with $\int_1^\infty f(t) dt = \infty$. Then the sequence is eventually Sylvester if and only if, for some integer $B \geq 0$,*

$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

At a late record, the summand is $(d_n - B)_+ f(U_n)$, where $d_n = U_{n+1} - U_n$. For $B = 0$, the integral comparison

$$\int_{R_n}^{R_{n+1}} f(t) dt \leq (R_{n+1} - R_n) f(U_n) \leq d_n f(U_n)$$

already charges for every newly reached height. The arithmetic is needed to retain divergence after subtracting a fixed B at each record. We shall find a progression of heights whose first crossings still cost at least one unit of excess jump.

Both $f(t) = 1/t$ and $f(t) = 1/[t \log(et)]$ are allowed; $1/t^2$ has a finite integral and is excluded. The relative limit $V_n/U_n \rightarrow 0$ does not supply the required summability.

Proof. Suppose the sequence is not eventually Sylvester. Absorption and integrality give

$$\frac{1}{U_n} \leq \frac{|V_n|}{U_n} = \frac{|E_n|}{C_n} \rightarrow 0.$$

Thus $U_n \rightarrow \infty$, so there are infinitely many record steps. At every sufficiently late one, $\rho_n = 1$, so a_n is coprime to L_n . The corresponding terms a_n are pairwise coprime: an

earlier term divides the later L_n , while the term at that record is coprime to L_n . For any fixed $B \geq 1$, choose B such multipliers $m_0, \dots, m_{B-1} > B$ and take T after their indices and after the threshold beyond which records have $\rho_n = 1$. Their size follows from $a_n \rightarrow \infty$.

Put $P = \prod_i m_i$ and choose x with $m_i \mid x + i$ by the Chinese remainder theorem. We select the heights $\tau = x + B + kP > R_T, k \in \mathbb{Z}$. They are spaced P apart, and the first is at most $R_T + P$.

At the first crossing of a selected height, $U_n \leq R_n < \tau \leq U_n + d_n$. In particular this is a record step. The full jump matters: if the previous record is 10 and a step goes from 5 to 12, then it first crosses 11, but its starting point is 5, not 10. The jump is 7, although the record increases by only 2. These numbers illustrate the distinction between the two increments; no denominator sequence satisfying (1) is asserted. If $d_n \leq B$, then $U_n \in [\tau - B, \tau)$, so

$$U_n = \tau - B + i = x + kP + i \quad \text{for some } 0 \leq i < B.$$

The choice of x gives $m_i \mid U_n$. Since $m_i \mid L_n$ as well, we obtain $m_i \mid d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \leq B < m_i$.

It follows that every crossing has $d_n > B$. A single jump may first cross several selected heights, so we must bound their combined weight by that jump's excess. If one step first crosses $h \geq 1$ selected heights, their spacing gives $(h - 1)P < d_n$. Write $r = d_n - B \geq 1$. Since $P \geq B + 1$, $d_n = B + r \leq Pr$, and hence $h \leq r$. Each of those heights exceeds U_n , so monotonicity gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each selected height above R_T has one first crossing. The heights are P apart, with the first at most $R_T + P$. For $R_N \geq R_T + P$, the intervals $[\tau, \min\{\tau + P, R_N\}]$ with $\tau \leq R_N$ cover $[R_T + P, R_N]$. Monotonicity bounds each integral by $Pf(\tau)$, so the crossing bounds give

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T + P}^{R_N} f(t) dt. \quad (11)$$

Since $R_n \rightarrow \infty$, the right side diverges for every $B \geq 1$; $B = 0$ follows by domination. Conversely, the eventual Sylvester recurrence gives by telescoping $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. $\square$

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (11) is $P^{-1} \log(\log(eR_N)/\log(e(R_T + P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Taking positive parts cannot increase the absolute difference. Since $V_n = U_n\theta_n$, the two nonnegative summands $U_nf(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+f(U_n)$ differ by at most $3U_nf(U_n)/a_n$. We verify that this comparison error is summable. Since $a_n x_n \rightarrow 1$, we have $C_n/a_n \sim qP_n/a_n^2$, and

$$\frac{P_{n+1}/a_{n+1}^2}{P_n/a_n^2} = \frac{a_n^3}{a_{n+1}^2} \rightarrow 0.$$

The ratio test gives $\sum_n C_n/a_n < \infty$; now $U_n \leq C_n$ and $f(U_n) \leq f(1)$ make the comparison error summable. The errors in (9) and (10) are also absolutely summable, since each is eventually between zero and $qP_n/a_{n+1} = O(P_n/a_n^2)$. Consequently Theorem 6.1 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_nf(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (12)$$

for some B . It remains to prove finiteness under the unrestricted hypotheses. The jump in either expression is $U_{n+1} - U_n$. Substituting $R_{n+1} - R_n$ loses the recovery from earlier decreases, which the crossing estimate uses. Nor would termwise convergence of the summands to zero establish finiteness.

Corollary 6.2 (a bound using the least common multiple). *Assume the hypotheses of Problem 1.1. Write $A_n = \text{lcm}(a_1, \dots, a_{n-1})$ with $A_1 = 1$. If*

$$\limsup_{n \rightarrow \infty} \frac{A_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < \infty,$$

then the sequence is eventually Sylvester.

Proof. Since $L_n/A_n = q/\gcd(q, A_n)$ lies in $[1, q]$, an upper bound on the expression in the corollary gives an upper bound on $(L_n/a_n)(a_n^2/a_{n+1} - 1)$; negative terms remain negative. Equation (10) therefore bounds the negative part of V_n . Choose an integer B above that eventual bound. The record series then vanishes after a finite prefix, so Theorem 6.1 applies. $\square$

The expression in Erdős and Straus's Theorem 3 is

$$\frac{[a_1, \dots, a_k]}{a_{k+1}} \left(\frac{a_{k+1}^2}{a_{k+2}} - 1 \right),$$

with a nonpositive upper limit [1, p. 132]. Setting $n = k + 1$ gives exactly $(A_n/a_n)\gamma_n$: the prefix ends before a_n , and the growth ratio begins at a_n . Tijdeman and Yuan extend this type of criterion to positive numerators [4]. Here any finite upper bound suffices

under the quadratic-limit assumption. Since $A_n \mid P_n$, the LCM hypothesis is no stronger than the product hypothesis in Corollary 4.9: if the product expression is at most B , the LCM expression is at most $\max(B, 0)$. When the earlier terms are pairwise coprime the two weights agree; repeated prime factors can make the LCM much smaller. This comparison does not assert the existence of a non-Sylvester rational example satisfying one bound but not the other.

The finite upper limit means an eventual upper bound; it does not require q to divide the LCM of the preceding denominators. For positive integer summand numerators b_n , the modified error is $V_n = b_n L_n - (a_n - 1)U_n$. The companion, Section 6 under “Integer coefficients”, separately proves boundedness from a lower bound on V_n and eventual constancy when $V_n/U_n \rightarrow 0$ is also assumed. The latter conclusion must not be read into the first hypothesis alone. The same section links finite examples separating them; compare Badea [3, p. 316] and Tijdeman–Yuan [4]. The companion’s Section 3 compares Duverney’s signed-series criterion [2, Corollary 3.2, p. 287], whose printed condition is one-sided, with the stronger absolute-convergence specialisation used there. It also records the different growth hypotheses for irrationality exponents in [14, Theorem 1, author-version p. 2]. Neither comparison is an input to the crossing theorem.

7 Further questions

Proposition 7.1 (necessary profile). *The integer tail of a sequence satisfying Problem 1.1’s hypotheses but not its conclusion has $E_n \neq 0$ eventually, $|E_n|/C_n \rightarrow 0$, unbounded negative magnitudes along negative indices, and*

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

Proof. Absorption excludes late zeros, descent excludes an eventually nonnegative error, and Theorems 4.1 and 5.1 exclude the two finiteness conditions. $\square$

For $B \geq 0$ put

$$F_B(X) = \sum_{\substack{n \in \mathcal{R} \\ U_n \leq X}} (-V_n - B)_+.$$

The weighted criterion reduces the problem to a statement about this cumulative sum. We would need to show, from growth and rationality, that for some fixed integer $B \geq 0$,

$$\liminf_{X \rightarrow \infty} \frac{F_B(X)}{X} = 0. \quad (13)$$

If the recurrence is not eventually Sylvester, first crossings instead give $F_B(X) \geq X/P_B - O_B(1)$, with an orbit-dependent CRT modulus P_B . The sum groups each record step by its starting numerator U_n , not by the running maximum R_n or the time index. If the eventual Sylvester recurrence fails, $U_n \rightarrow \infty$, so only finitely many of these steps begin below any fixed X . To see the equivalence with an admissible weight, enumerate the record indices by j and apply the next lemma with $u_j = U_n$ and $w_j = (-V_n - B)_+$. The lemma supplies a weight once (13) holds. Establishing that estimate would contradict the first-crossing lower bound and exclude every non-Sylvester rational tail.

Lemma 7.2 (weights and linear density). *Let u_j be positive integers, let $w_j \geq 0$, and put $F(X) = \sum_{u_j \leq X} w_j$, with the sum allowed a priori to be $+\infty$. Then $\liminf_{X \rightarrow \infty} F(X)/X = 0$ if and only if there is a finite nonincreasing $f : [1, \infty) \rightarrow [0, \infty)$ with $\int_1^\infty f(t) dt = \infty$ and $\sum_j w_j f(u_j) < \infty$.*

Proof. Suppose the lower limit vanishes. Choose cutoffs at which the ratio $F(X)/X$ is small. Since F is constant on each $[m, m+1)$, there are integers $X_1 < X_2 < \dots$ with $X_k \geq 2^k$ and $F(X_k) \leq 2^{-k} X_k$. Set

$$f(t) = \sum_{k \geq 1} \frac{\mathbf{1}_{[1, X_k]}(t)}{X_k}.$$

This function is nonincreasing and bounded by $\sum_k 2^{-k}$. Each cutoff contributes $1 - 1/X_k$ to its integral, but at most 2^{-k} to the weighted sum. Interchanging nonnegative sums therefore gives

$$\int_1^\infty f(t) dt = \sum_k (1 - 1/X_k) = \infty, \quad \sum_j w_j f(u_j) = \sum_k \frac{F(X_k)}{X_k} \leq 1.$$

Conversely, let f be such a weight and suppose $F(m) \geq cm$ for some $c > 0$ and all integers $m \geq M$. A nonincreasing nonnegative function with divergent integral is positive everywhere, so $F(X) \leq f(X)^{-1} \sum_j w_j f(u_j)$ is finite. Partial summation gives, for $N > M$,

$$\sum_{u_j \leq N} w_j f(u_j) = F(N)f(N) + \sum_{m=1}^{N-1} F(m)(f(m) - f(m+1)) \geq c(Mf(M) + \sum_{m=M+1}^N f(m)),$$

every term being nonnegative. The right side diverges as $N \rightarrow \infty$, because $\sum_{m \geq 1} f(m) \geq \int_1^\infty f(t) dt = \infty$. $\square$

7.1 Arithmetic compatibility

The remaining estimate requires the denominator recurrence as well as control of the error. For example, $C_n = n^2 + 1$, $E_n = -(2n + 1)$ has vanishing relative error and the necessary scalar pattern. But at $n = 2, 3, 4$ its numerators are 5, 10, 17. The update $10 = 5a_2 - D_2$ forces $5 \mid D_2$, hence $5 \mid D_3 = a_2 D_2$; the next update would force $5 \mid 17$. Thus these numerators cannot satisfy (1). The companion's Section 14 also constructs sequences avoiding sparse fixed moduli and distinguishes that avoidance from compatibility with all later denominators. Neither property alone supplies (13).

Conversely, suppose integer sequences indexed from zero satisfy (1), with $a_n > 1$, $C_n > 0$, $D_n \geq 0$ and $E_n/C_n \rightarrow 0$. Section 4 of the companion proves that $D_n > 0$, $C_n/D_n \rightarrow 0$, $\sum_n 1/a_n = C_0/D_0$ and $a_{n+1}/a_n^2 \rightarrow 1$. All these hypotheses concern the infinite sequences; a scalar example or a finite initial segment satisfies only some of them.

Problem 7.3 (growth of repeated denominator factors). For every rational-tail orbit satisfying the hypotheses but not the conclusion of Problem 1.1, must

$$\limsup_{n \rightarrow \infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a $K \geq 1$ for which

$$2^n \leq M_n^K$$

at infinitely many indices?

Section 2 gives $C_{n+1}/C_n \rightarrow 1$; averaging logarithms therefore gives $\log C_n = o(n)$. Since $1 \leq M_n \leq C_n$, every such orbit already satisfies $\log M_n/n \rightarrow 0$. A positive answer therefore holds exactly when no such orbit exists: proving it from failure of Sylvester behaviour would exclude every counterexample.

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A A factorial modulus for integral recursion

A constant-negative error $E_n = -m$ with $C_0 = c$ gives $C_n = c + nm$ and the identity

$$D_n + m = (a_n - 1)C_n. \quad (14)$$

In the case $m = c = 1$, where (14) reads $D_n + 1 = (a_n - 1)(n + 1)$, each multiplier is determined by its predecessor. The recursion is $a_{n+1} = \text{num}(n, a_n)/(n + 2)$, where

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the [forced numerator](#). We stop at the first nonintegral quotient. This procedure checks divisibility only; it does not impose the positivity conditions required of an infinite reciprocal-tail sequence. The [formal exact-division condition](#) requires exact division at each step.

Example A.1 (a nonintegral second step). Starting from $a_0 = 3$, the first numerator $\text{num}(0, 3) = 6$ is divisible by 2, giving $a_1 = 3$. The next numerator $\text{num}(1, 3) = 13$ is not divisible by 3, so the recursion stops.

Direct iteration can produce very large intermediate integers. To decide whether the first h divisions are exact, however, it suffices to know the initial value modulo $(h + 1)!$. Computing a pseudo-greedy orbit through residues modulo a shrinking product modulus is Koizumi's method [10, Remark 2 and Algorithm 1, pp. 13–14]; for the forced numerator that product is a factorial.

Theorem A.2 (factorial residue reduction). *Let h be a nonnegative integer and let a, b be integers with $a \equiv b \pmod{(h + 1)!}$. The first h steps of the recursion $a_{n+1} = \text{num}(n, a_n)/(n + 2)$, starting at index zero, are integral for $a_0 = a$ if and only if they are integral for $a_0 = b$.*

Proof. Define $M(0, i) = 1$ and $M(h + 1, i) = (i + 2)M(h, i + 1)$. Thus $M(h, 0) = (h + 1)!$. We prove the stronger statement that, at index i , congruent inputs modulo $M(h, i)$ give the same answer to whether the next h quotients are all integral. There is nothing to prove for $h = 0$.

For the inductive step, suppose $a \equiv b \pmod{(i+2)M(h, i+1)}$. Since $\text{num}(i, \cdot)$ is an integer polynomial, its values at a and b are congruent modulo that product. In particular, one is divisible by $i+2$ if and only if the other is. If neither is divisible, both recursions stop. Otherwise their quotients are congruent modulo $M(h, i+1)$, so the induction hypothesis applies to the remaining h updates at index $i+1$. Taking $i=0$ proves the claim. $\square$

For $h=1$, the condition is $2 \mid a^2 - 2a + 3$, or equivalently that a is odd. This illustrates Theorem A.2 with modulus $2! = 2$.

The enumeration in Example 9.1 of the companion, over $2 \leq a_0 < 5000$, gives at most 17 successful updates (18 values including a_0). The seed $a_0 = 1$ is excluded because it is fixed and violates $a_n \geq 2$. This finite calculation establishes only the stated range. The infinite [constant-negative exclusion](#) requires the separate argument under the hypotheses given there.

B Sources and related work

The companion *Reciprocal Sums and the Sylvester Recurrence: Further Results and Proofs* gives full proofs and fixed-revision source links. [Section 2](#) treats regular rates and the stronger cubic disagreement result; [Sections 4–5](#) develop the integer tail numerators. The gcd and bounded-increment arguments culminate in [Section 12](#). The LCM results are in [Section 6](#), scalar summability in [Section 13](#), and open estimates and failed constructions in [Section 14](#). A final map records the dependencies. The auxiliary constructions need not be rational reciprocal series; they satisfy only the hypotheses stated for them.

The older integer-remainder method is represented by Erdős–Straus [12, Theorem 2.1, pp. 85–86]. The polynomial Cantor-series hypotheses in Hančl–Tijdeman [13, Theorem 2.2, pp. 39–40] and the freely selectable series of Problem #270 [15, Theorems 1–2] concern different settings; the companion’s [Section 3](#) explains the distinction. The Isabelle/HOL work [16] formalises older Erdős–Straus criteria, not Problem #243. The general regular-rate extraction in the companion also uses the standard Gamma-ratio asymptotics recorded in [17]; the cubic specialisation here uses only its explicit polynomial F_n .

Indexing of the canonical tails. The construction in [Section 2](#) has $C_1 = p$, $D_1 = q$. For a zero-based recurrence orbit, set $a_0 = 1$, $D_0 = q$ and $C_0 = p + q$; one update gives those initial values. This extends the two recurrences, but supplies neither $a_0 > 1$ nor strict increase at that first step. For the zero-based strictly increasing sequence in the formal cubic theorem, instead choose N with $a_N > N$ and put $b_n = n + 1$ for $0 \leq n < N$, $b_n = a_n$ for $n \geq N$. Then (b_n) is strictly increasing, its reciprocal sum differs from the original sum by a rational finite sum, and its rate has the same index n after N . In particular, no shift of $3/n$ is made. The companion’s [indexing subsection](#) gives this finite-prefix bridge.

C Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The links identify fixed-revision Lean declarations and recorded Comparator checks. The printed cubic argument uses Chebotarev; the formal square-specialisation proof uses a Dedekind-zeta pole comparison. Their common conclusion does not check each step of the printed proof. This revision includes no new Lean build, Comparator replay or independent review. The evidence record retains source-pin and index gaps, and the estimate in Section 7 remains open.

The formal [factorial residue reduction](#) uses the [shrinking-modulus induction](#), [polynomial congruence](#), and [cancellation after exact division](#). Its modulus is identified by the [ascending-factorial formula](#) and its factorial value at the initial index.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.2	Lean · Comparator	Proposition 3.1	Lean · Comparator
Theorem 4.1	Lean · Comparator	Proposition 4.2	Lean · Comparator
Theorem 4.3	Lean · Comparator	Corollary 4.4	Lean · Comparator
Proposition 4.5	Lean · Comparator	Lemma 4.6	Lean · Comparator
Theorem 4.7	Lean · Comparator	Corollary 4.8	Lean · Comparator
Corollary 4.9	Lean · Comparator	Corollary 4.10	Lean · Comparator
Theorem 5.1	Lean · Comparator	Theorem 6.1	Lean · Comparator
Corollary 6.2	Lean · Comparator	Proposition 7.1	Lean · Comparator
Lemma 7.2	Lean · Comparator	Theorem A.2	Lean · Comparator